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import numpy as np

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_squared_error, r2_score


X = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10]).reshape(-1, 1)
y = np.array([10, 15, 20, 25, 30, 35, 40, 45, 50, 55])


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Linear Regression")
print("-----")
print("Actual Values:", y_test)
print("Predicted Values:", y_pred)
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)
new_value = np.array([[12]])
prediction = model.predict(new_value)
print("Prediction for 12:", prediction[0])
plt.scatter(X, y)
plt.plot(X, model.predict(X))
plt.xlabel("X")
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plt.ylabel("Y")  
plt.title("Linear Regression")  
plt.show()
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... Linear Regression
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Actual Values: [50 15]
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Predicted Values: [50. 15.]
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Mean Squared Error: 1.5777218104420236e-30
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R2 Score: 1.0
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Slope: 5.000000000000001
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Intercept: 4.9999999999999964
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Prediction for 12: 65.00000000000001
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Linear Regression

